# CH 4 - Inverses and Determinants

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## Matrix Inverses

Recall that $I = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 \end{bmatrix} = [\vec{e}_1 \ \vec{e}_2 \ \dots \ \vec{e}_n]$

Also $AI = IA = A$ for all $m \times n$ matrices A

🐳 **Info — Left and Right Inverses**

Let $A$ be a $m \times n$ matrix.

If $B$ is a $n \times m$ matrix s.t $AB = I_m$, then $B$ is called a **right inverse** of $A$.

$A$ has a right inverse if and only if rank $A = m$

If $C$ is a $n \times m$ matrix s.t $CA = I_m$, then $C$ is called a **left inverse** of $A$.

$A$ has a left inverse if and only if rank $A = n$

From above, we can obtain $A\left[\vec{b}_1 \ \dots \ \vec{b}_n\right] = I$

$\implies [A|\vec{e}_i], \forall i = 1, \dots, m$ is consistent

$\implies \vec{e}_i \in \text{Col}(A), \forall i = 1, \dots, m.$

Thus $\text{Span}\{\vec{e}_1, \dots, \vec{e}_n\} \subseteq \text{Col}(A) = \mathbb{R}^m \implies \text{rank}(A) = m$

🐳 **Info — Rank-Inverse**

If $A$ is a $m \times n$ matrix, then

1. $A$ has a right inverse if and only if $\text{rank}(A) = m$
2. $A$ has a left inverse if and only if $\text{rank}(A) = n$

For $A$ be $n \times n$, a square matrix, it has both a left and right inverse. Those inverses are identical.

$A$, a square matrix, is also invertible if and only if $\text{rank}(A) = n$

🐳 **Info — Matrix Inverse Invertible**

Let $A$ be an $n \times n$ matrix. If $B$ is a matrix s.t. $AB = I = BA$, then $B$ is called the **inverse** of $A$. We write $B = A^{-1}$ and we say that $A$ is invertible.

$B$ is unique.

🐳 **Info — Properties of Invertible Matrices**

If $A, B$ are invertible matrices and $c \in \mathbb{R}$ with $c \neq 0$, then

1. $(cA)^{-1} = \frac{1}{c}A^{-1}$
2. $\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$
3. $(AB)^{-1} = B^{-1}A^{-1}$

**Proof**

Property 3

$(AB)B^{-1}A^{-1} = A(BB^{-1})A^{-1} = AIA^{-1} = IAA^{-1} = II = I$

So $(AB)^{-1} = B^{-1}A^{-1}$

$\square$

To find the inverse of an invertible matrix, we note that the following $n$ linear systems must be consistent:

$[A \mid \vec{e}_1], [A \mid \vec{e}_2], \ldots, [A \mid \vec{e}_n]$.

The solutions to these systems will be the columns of $A^{-1}$

Since $A$ is the same coefficient matrix in each system, it will be the same sequence of elementary row operations to go from $A$ to its RREF in each system.

Thus we can create a multiple augmented matrix: $[A|\ \vec{e}_1\ \ldots\ \vec{e}_n] = [A \mid I]$, then row reduce to $[I \mid B]$, where $B = A^{-1}$

Example:

$[A \mid I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{cc|cc} 1 & 2 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$

$A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$

🐳 **Info — General Forumla for $M_{2\times 2}$**

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2\times 2}(\mathbb{R})$. Then $A$ is invertible if and only if $ad - bc \neq 0$. Moreover, if $A$ is invertible

$$A^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

🐳 **Info — Invertible Matrix Theorem**

For any $n \times n$ matrix $A$, the following are equivalent:

1. $A$ is invertible
2. The RREF of $A$ is $I$
3. rank $A = n$
4. The system of equations $A\vec{x} = \vec{b}$ is consistent with a unique solution $\forall b \in \mathbb{R}^n$
5. The nullspace of $A$ is $\left\{\vec{0}\right\}$
6. THe columns of $A$ form a basis for $\mathbb{R}^n$
7. The rows of $A$ form a basis for $\mathbb{R}^n$
8. $A^T$ is invertible

## Elementary Matrices

🐳 **Info — Elementary Matrix**

An $n \times n$ matrix $E$ is called an **elementary matrix** if it can be obtained from the $n \times n$ identity matrix by performing exactly one elementary row operation.

- If $E$ is an elementary matrix, then $E$ is invertible and $E^{-1}$ is also an elementary matrix
- If $A$ is an $m \times n$ matrix and $E$ is the $m \times m$ elementary matrix corresponding to the row operation $R_i + cR_j$ or $cR_i$ or $R_i \leftrightarrow R_j, i \neq j$ respectively, then $EA$ is the matrix obtained from $A$ by performing the row operation $R_i + cR_j$ or $cR_i$ or $R_i \leftrightarrow R_j$ respectively on $A$.
  - also

    $$\text{rank } (EA) = \text{rank } A$$

Examples:

1. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \sim 5R_1 \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$
2. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \sim R_1 \leftrightarrow R_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
3. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \sim R_1 + 7R_2 \begin{bmatrix} 1 & 0 \\ 7 & 1 \end{bmatrix}$
4. Consider $E = \begin{bmatrix} 1 & 0 \\ 7 & 1 \end{bmatrix}$ ia an elementary matrix, corresponding to the row operation $R_2 + 7R_1$. The elementary matrix correspond to $R_2 - 7R_1$ is mat(1, 0, −7, 1) is $E^{-1}$

   $\begin{bmatrix} 1 & 0 \\ 7 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -7 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

**Proof**

Visual aid:

$$E = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & c & \dots & 1 \end{bmatrix}$$

Row $i$ has $e_{ii} = 1$, $e_{ij} = c$ and everywhere else is 0

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2k} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{j1} & a_{j2} & \dots & a_{jk} & \dots & a_{jn} \\ a_{i1} & a_{i2} & \dots & a_{ik} & \dots & a_{in} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mk} & \dots & a_{mn} \end{bmatrix}$$

🐳 **Info — Matrix Decomposition**

If $A$ is a $m \times n$ matrix with RREF $R$, then there exists a sequence $E_1, \dots, E_k$ of $m \times m$ elementary matrices s.t. $E_k \dots E_2 E_1 A = R$. In partticular $A = E_1^{-1} E_2^{-1} \dots E_k^{-1} R$

Note that the order of $E_i$ matters

- If $A$ is a $n \times n$ invertible matrix, then $A, A^{-1}$ can be written as a product of elementary matrices.

If $E$ is an elementary matrix, its transpose is also an elementary matrix

## Determinants

🐳 **Info — Determinants**

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. The **determinant** of $A$ is

$$\det\ A = ad - bc$$

Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. The **determinant** of $A$ is

$$\begin{aligned} \det\ A &= a \det \begin{bmatrix} e & f \\ h & i \end{bmatrix} - d \det \begin{bmatrix} b & c \\ h & i \end{bmatrix} + g \det \begin{bmatrix} b & c \\ e & f \end{bmatrix} \\ &= a(ei - fh) - d(bi - ch) + g(bf - ce) \end{aligned}$$

Let $A$ be a $n \times n$ matrix with $n \geq 2$.

1. The **determinant** of $A$ is defined

$$\det\ A = \sum_{i=1}^{n} a_{i1} C_{i1}$$

   where the determinant of a $1 \times 1$ matrix is defined by $\det[c] = c$

2. For any $1 \leq i \leq n$

$$\det\ A = \sum_{k=1}^{n} a_{ik} C_{ik}$$

   called thee **cofactor expansion across the i-th row**,

3. For any $1 \leq j \leq n$

$$\det\ A = \sum_{k=1}^{n} a_{kj} C_{kj}$$

   called **cofactor expansion across the j-th column**

Examples:

1. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\det\ A = 1 \cdot 4 - 2 \cdot 3 = -2$
2. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

   $\det\ A = 1(-1)^{1+1}\ \det\ A(1,1) + 4(-1)^{2+1}\ \det\ A(2,1) + 7(-1)^{3+1}\ \det\ A(3,1) = 0$

🐳 **Info — Cofactor**

Let $A$ be a $n \times n$ matrix with $n \geq 2$. Let $A(i,j)$ e then $(n-1) \times (n-1)$ matrix obtained from $A$ by deleting the $i$-th row and $j$-th column. The **cofactor** of $a_{ij}$ is

$$C_{ij}(A) = (-1)^{i+j}\ \det\ A(i,j)$$

Examples:

1. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

   $A(1,1) = \begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix}, A(3,2) = \begin{bmatrix} 1 & 3 \\ 4 & 6 \end{bmatrix}$

   $C_{11} = (-1)^{1+1}\ \det\ A(1,1) = 5 \cdot 9 - 6 \cdot 8 = -3$

   $C_{32} = (-1)^{3+2}\ \det\ A(3,2) = -(6-12) = 6$
2. $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 6 \\ 5 & -2 & 0 \end{bmatrix}$

   $\det\ A = a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33} = 0 + 6C_{23} + 0 = 6(-1)^{2+3}(-2-10) = 72$

🐳 **Info — Triangular**

A $m \times n$ matrix $U$ is said to be **upper triangular** if $u_{ij} = 0$ whenever $i > j$.

A $m \times n$ matrix $L$ is said to be **lower triangular** if $l_{ij} = 0$ whenever $i < j$

If a $n \times n$ matrix $A$ is an upper or lower triangular, then

$$\det\ A = a_{11}a_{22}a_{33}...a_{nn}$$

Examples:

1. Compute $\det\ A$

   Let $A = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 3 & 5 & 2 & 0 & 0 \\ 6 & 2000 & 5 & 1 & 0 \\ \pi & e & 7 & -3 & 5 \end{bmatrix}$

   $\det\ A = 2(-1)^{1+1}\ \det \begin{bmatrix} -1 & 0 & 0 & 0 \\ 5 & 2 & 0 & 0 \\ 2000 & 5 & 1 & 0 \\ e & 7 & -3 & 5 \end{bmatrix} = 2(-1)\ \det \begin{bmatrix} 2 & 0 & 0 \\ 5 & 1 & 0 \\ 7 & -3 & 5 \end{bmatrix} = (2)(-1)(2)\ \det \begin{bmatrix} 1 & 0 \\ -3 & 5 \end{bmatrix} =$

   $(2)(-1)(2)(5) = -20$

2. Compute $\det\ A$

   Let $A = \begin{bmatrix} 2 & 0 & 6 \\ -1 & 5 & 3 \\ 1 & 2 & 1 \end{bmatrix}$.

   We expand across the first row:

   $\det\ A = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$

   $$= 2(-1)^{1+1}\ \det\begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix} + 0 + 6(-1)^{3+1}\ \det\begin{bmatrix} -1 & 5 \\ 1 & 2 \end{bmatrix} = -2 - 42 = -44$$

3. Compute $\det\ B$

   Let $B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 306 & 7 & 0 & 0 \\ 812 & \pi & -1 & 0 \\ 4 & -2 & 18 & 5 \end{bmatrix}$

   $\det\ B = (1)(7)(-1)(5) = -35$

🐳 **Info — Determinants and Row Operations**

1. Row multiplication
   If $A$ is a $n \times n$ matrix and $B$ is the matrix obtained from $A$ by multiplying one row of $A$ by $c \in \mathbb{R}$, then

   $$\det\ B = c\ \det\ \mathrm{A}$$

2. Row swap

   If $A$ is a $n \times n$ matrix and $B$ is the matrix obtained from $A$ by swaping one row of $A$ to another, then

   $$\det\ B = -\ \det\ \mathrm{A}$$

   - If $A$ has two identical rows, then $\det\ A = 0$

3. If $A$ is a $n \times n$ matrix and $B$ is the matrix obtained from $A$ by adding a multiple of one row of $A$ to another, then

   $$\det\ B = \det\ \mathrm{A}$$

Concept of Proof for Row multiplication:

If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{32} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

We multiply $cR_2$ to $B \implies \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ ca_{21} & ca_{22} & ca_{32} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$\det\ B = C_{21}b_{21} + C_{22}b_{22} + C_{32}b_{32} = c(C_{21}a_{21}, C_{22}a_{22}, C_{32}a_{32}) = c\ \det\ A$

Example:

$$\det \begin{bmatrix} 2 & 1 & 7 \\ 3 & 1 & 2 \\ 1 & -1 & 0 \end{bmatrix} R_1 \leftrightarrow R_2 = -\det \begin{bmatrix} 1 & -1 & 0 \\ 3 & 1 & 2 \\ 2 & 1 & 7 \end{bmatrix} R_2 - 3R_1, R_3 - 2R_1$$

$$= -\det \begin{bmatrix} 1 & -1 & 0 \\ 0 & 4 & 2 \\ 0 & 3 & 2 \end{bmatrix} \tfrac{1}{4}R_2 = -4 \det \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 3 & 7 \end{bmatrix} R_1 + R_2, R_3 - 3R_2 = -4 \det \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & \frac{11}{2} \end{bmatrix} = -22$$

Or you can stop at step at $-\det \begin{bmatrix} 1 & -1 & 0 \\ 0 & 4 & 2 \\ 0 & 3 & 2 \end{bmatrix}$ to use cofactor by taking advantage of the zeros in the first column.

🐳 **Info — Determinants and Elementary Matrices**

1. If $A$ is a $n \times n$ matrix and $E$ is an $n \times n$ elementary matrix, then
$$\det\ EA = \det E\ \det A$$
2. A $n \times n$ matrix $A$ is invertible if and only if $\det\ \mathrm{A} \neq 0$
3. If $A, B$ are $n \times n$ matrices, then $\det(AB) = \det A\ \det B$
4. If $A$ is an invertible matrix, then $\det\ A^{-1} = \frac{1}{\det A}$
5. If $A$ is a $n \times n$ matrix, $\det A = \det\ A^T$

Concept for Proof of theorem 1 from above

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} cR_2 \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix} c \Longrightarrow \det\ E = c$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_2 \leftrightarrow R_1 \sim \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} c \Longrightarrow \det\ E = -1$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_1 + cR_2 \sim \begin{bmatrix} 1 & c & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} c \Longrightarrow \det\ E = 1$$

## Determinants and Systems of Equations

Consider $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ also invertible.

Compute all possible cofactors:

$C_{11} = (-1)^{1+1}\ \det[d] = d$

$C_{12} = (-1)^{2+1}\ \det[c] = -c$

$C_{21} = (-1)^{2+1}\ \det[b] = -b$

$C_{22} = (-1)^{2+2}\ \det[a] = a$

Consider a matrix of cofactors:

$\mathrm{Cof}(A) = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$

Notice that $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{\det A}\ \mathrm{Cof}(A)$

🐳 **Info — Inverse by Cofactors**

1. If $A$ is a $n \times n$ matrix with cofactors $C_{ij}$ and $i \neq j$, then

$$\sum_{k=1}^{n} (A)_{ik} C_{jk} = 0$$

   Note that this is different from the way of computing the determinant due to the index $j$

2. If $A$ is a $n \times n$ matrix, then

$$(A^{-1})_{ij} = \frac{1}{\det A} C_{ji}$$

🐳 **Info — Cofactor Matrix and Adjugate**

Let $A$ is a $n \times n$ matrix. The **cofactor matrix**, cof $A$, of $A$ is the matrix whose entry is the $ij-$th cofactor of $A$.

$$(\text{cof} A)_{ij} = C_{ij}$$

The **adjugate** of $A$ is the matrix defined by

$$(\text{adj } A)_{ij} = C_{ji}$$

In particular, adj $A = (\text{cof} A)^T$

Example:

Let $A = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$

$\det A = (-1)(-1)^{1+1} \det \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} + 0 + 1(-1)^{3+1} \det \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} = -1 + -2 = -3$

$C_{11} = 1$

$C_{12} = (-1)^{2+1} \det \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} = 4$

$C_{13} = (-1)^{3+1} \det \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} = -2$

$C_{21} = (-1)^{2+1} \det \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix} = -1$

$C_{22} = (-1)^{2+2} \det \begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix} = -5$

$C_{23} = (-1)^{2+3} \det \begin{bmatrix} -1 & 0 \\ 2 & 1 \end{bmatrix} = 1$

$C_{31} = (-1)^{3+1} \det \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} = -1$

$C_{32} = (-1)^{3+2} \det \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix} = 2$

$C_{33} = (-1)^{3+3} \det \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -1$

$A^{-1} = -\frac{1}{3} \begin{bmatrix} 1 & 4 & -2 \\ -1 & -5 & 1 \\ -1 & 2 & -1 \end{bmatrix}$

**Cramer's Rule**

Let $A = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$

Let $\vec{b} = \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix}$

Solve $A\vec{x} = \vec{b}$

Since $A$ is invertible, we will left-multiply both sides by $A^{-1}$

$A^{-1}A\vec{x} = A^{-1}\vec{b}$

$\vec{x} = A^{-1}\vec{b}$

$$= -\frac{1}{3}\begin{bmatrix} 1 & 4 & -2 \\ -1 & -5 & 1 \\ -1 & 2 & -1 \end{bmatrix}\begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & -2 \\ -1 & -5 & 1 \\ -1 & 2 & -1 \end{bmatrix}\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

## CUTOFF QUIZ